

1	(a)	$(FH =) \sqrt{12^2 + 12^2} (= 16.97... \text{ or } \sqrt{288} \text{ or } 12\sqrt{2})$	3	M1
		$\tan CFH = \frac{10}{16.97...} \text{ oe}$ or e.g. $(CF =) \sqrt{16.97^2 + 10^2} (= 19.69... \text{ or } \sqrt{388} \text{ or } 2\sqrt{97})$ and e.g. $\frac{\sin CFH}{10} = \frac{\sin 90}{16.97}$		M1 for a correct trig statement involving CFH
			30.5	A1 accept 30.4 – 30.7
	(b)	$(BG =) 10 + \sqrt{15^2 - 12^2} (= 19)$	3	M1
		$(BE =) \sqrt{19^2 + 16.97...^2} \text{ oe}$		M1 fit their FH
			25.5	A1 accept 25.4 – 25.6
Total 6 marks				

2		$(AH =) \sqrt{6^2 + 5^2 + 9^2} (= \sqrt{142}) \text{ or}$ $(FH = GE =) \sqrt{5^2 + 9^2} (= \sqrt{106})$ E.g. $\sin AHF = \frac{6}{\sqrt{142}} \text{ or } \tan AHF = \frac{6}{\sqrt{106}} \text{ or}$ $\cos AHF = \frac{\sqrt{106}}{\sqrt{142}} \text{ or}$ $\sin FAH = \frac{\sqrt{106}}{\sqrt{142}} \text{ or } \cos FAH = \frac{6}{\sqrt{142}} \text{ or}$ $\tan FAH = \frac{\sqrt{106}}{6}$	4	M1 for working out AH or FH or GE
		E.g. $\sin AHF = \frac{6}{\sqrt{142}} \text{ or } \tan AHF = \frac{6}{\sqrt{106}} \text{ or}$ $\cos AHF = \frac{\sqrt{106}}{\sqrt{142}} \text{ or}$ $\sin FAH = \frac{\sqrt{106}}{\sqrt{142}} \text{ or } \cos FAH = \frac{6}{\sqrt{142}} \text{ or}$ $\tan FAH = \frac{\sqrt{106}}{6}$		M1 for a correct method for finding angle AHF or finding angle FAH Allow $\cos AHF = \left(\frac{\sqrt{142}^2 + \sqrt{106}^2 - 6^2}{2 \times \sqrt{142} \times \sqrt{106}} \right) \text{ oe or}$ $\sin AHF = \frac{\sin 90}{\sqrt{142}} \times 6 \text{ oe}$
		E.g. $\sin^{-1} \left(\frac{6}{\sqrt{142}} \right) \text{ or } \tan^{-1} \left(\frac{6}{\sqrt{106}} \right)$ or $\cos^{-1} \left(\frac{\sqrt{106}}{\sqrt{142}} \right) \text{ or}$ $90 - \sin^{-1} \left(\frac{\sqrt{106}}{\sqrt{142}} \right) \text{ or } 90 - \cos^{-1} \left(\frac{6}{\sqrt{142}} \right)$ or $90 - \tan^{-1} \left(\frac{\sqrt{106}}{6} \right)$		M1 for a complete method Allow $\cos^{-1} \left(\frac{\sqrt{142}^2 + \sqrt{106}^2 - 6^2}{2 \times \sqrt{142} \times \sqrt{106}} \right) \text{ oe or}$ $\sin^{-1} \left(\frac{\sin 90}{\sqrt{142}} \times 6 \right) \text{ oe}$
			30.2	A1 for 30.2 – 30.3
Total 4 marks				

3		$[AM =] \sqrt{5^2 + 15^2} (= \sqrt{250} = 15.8...)$ where M is midpoint of EF , oe other correct method to find AM $[AD =] \sqrt{12^2 + 15^2} (= \sqrt{369} = 19.2...)$ $[DM =] \sqrt{12^2 - 5^2} (= \sqrt{119} = 10.9...)$	4	M2 for a complete method to find two of AM , AD , DM (where M is the midpoint of EF) Other longer ways to find AM , AD , DM may be used but must be a complete method eg $\angle DEM = \cos^{-1} \left(\frac{5}{12} \right) (= 65.37...)$ and $DM = 12 \sin 65.37...$ $\angle DEM = \cos^{-1} \left(\frac{5}{12} \right) (= 65.37...)$ and $DM = 5 \tan 65.37...$ Use $10 \div 2$ as 5 throughout (M1 For a complete method to find one of AM , AD , DM (where M is the midpoint of EF))
		eg $\tan DAM = \frac{\sqrt{119}}{\sqrt{250}} \left(= \frac{10.9...}{15.8...} \right) \text{ oe}$ or $\sin DAM = \frac{\sqrt{119}}{\sqrt{369}} \left(= \frac{10.9...}{19.2...} \right) \text{ oe}$ or $\cos DAM = \frac{\sqrt{250}}{\sqrt{369}} \left(= \frac{15.8...}{19.2...} \right) \text{ oe}$		M1 a correct method to find the required angle – other longer methods may be used but they must get to the stage of an equation for the required angle eg $\sin DAM = \frac{10.9...}{\sqrt{15.8...^2 + 10.9...^2}}$ NB: “10.9...” and “15.8...” must come from correct working
		Working not required, so correct answer scores full marks (unless from obvious incorrect working)	34.6	A1 any answer which rounds to 34.6
Total 4 marks				

7	$(AC =) \sqrt{8^2 + 18^2} (= \sqrt{388} = 2\sqrt{97} = 19.697\dots)$ or $(CE =) \sqrt{8^2 + 18^2 + 12^2} (= \sqrt{532} = 2\sqrt{133} = 23.065\dots)$ oe		3	M1
	eg $\tan ECA = \frac{12}{\sqrt{388}}$ or $\sin ECA = \frac{12}{\sqrt{532}}$ or $\cos ECA = \frac{\sqrt{388}}{\sqrt{532}}$ or $\sin ECA = \frac{\sin 90 \times 12}{\sqrt{532}}$ or $\cos ECA = \left(\frac{(\sqrt{388})^2 + (\sqrt{532})^2 - 12^2}{2 \times \sqrt{388} \times \sqrt{532}} \right)$ oe			M1 for a correct trig statement with ECA as the only unknown. NB allow use 'x' or other variable in place of ECA .
		31.4		A1 allow 31.3 – 31.5
Total 3 marks				

8	eg $(AM =) \sqrt{x^2 + (4x)^2} (= \sqrt{17x^2} = x\sqrt{17})$ oe or $(AM =) \sqrt{(0.5x)^2 + (2x)^2} (= \sqrt{\frac{17}{4}x^2} = x\sqrt{\frac{17}{4}})$ oe or $(AM =) \sqrt{20^2 + 5^2} (= \sqrt{425} = 5\sqrt{17})$ oe		5	M1 for a correct method to find AM as a numerical value or in algebraic form, must have brackets or recover
	Height of triangle eg $\sqrt{(2x)^2 - x^2} (= \sqrt{3x^2} = x\sqrt{3})$ oe or $\sqrt{x^2 - (0.5x)^2} (= \sqrt{\frac{3}{4}x^2} = x\sqrt{\frac{3}{4}})$ oe or $\sqrt{10^2 - 5^2} (= \sqrt{75} = 5\sqrt{3})$ oe			M1 for a correct method to find height of equilateral triangle HJK as a numerical value or in algebraic form
	eg $\tan MAJ = \frac{\sqrt{3}+2}{\sqrt{17}}$ or $\tan MAJ = \frac{\frac{\sqrt{3}}{2}+1}{\frac{\sqrt{17}}{2}}$ or $\tan MAJ = \frac{5\sqrt{3}+10}{5\sqrt{17}}$			M1 for correct values for the correct angle (no algebra) or for $\tan MAJ$ is given numerically in the range 0.9 – 0.91
	eg $\frac{(\sqrt{3}+2)}{\sqrt{17}} \times \frac{\sqrt{17}}{\sqrt{17}} (= \frac{\sqrt{51}+2\sqrt{17}}{17})$			M1
		$\frac{\sqrt{68}+\sqrt{51}}{17}$		A1 $\frac{\sqrt{51}+\sqrt{68}}{17}$
Total 5 marks				

9	$(AD =) \frac{2.2}{\tan 18}$ ($= 6.77\dots$) or $(EA =) \frac{2.2}{\sin 18}$ ($= 7.11\dots$)		4	M1 a correct method to find AD or AE
	$(DB =) \sqrt{("6.77\dots")^2 + 6^2}$ ($= 9.04\dots$) or $(EB =) \sqrt{6^2 + "7.11\dots"^2}$ ($= 9.31\dots$) or $(EB =) \sqrt{6^2 + "6.77\dots"^2 + 2.2^2}$ ($= 9.31\dots$)			M1 a correct method to find DB or EB
	$\tan DBE = \frac{2.2}{"9.04\dots"}$ or $\sin DBE = \frac{2.2}{"9.31\dots"}$ or $\sin DBE = \frac{2.2 \sin 90}{"9.31\dots"}$ $\cos DBE = \frac{"9.04\dots"}{"9.31\dots"}$ or use of cosine rule			M1 complete method to find one of $\tan DBE$ or $\sin DBE$ or $\cos DBE$ —NB: if using cosine, the student will need to have found DB and EB previously
	Correct answer scores full marks (unless from obvious incorrect working)	13.7		A1 Allow answers in range 13.59 – 13.8
Total 4 marks				

10	$[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe [where } N \text{ is the midpoint of } EC]$ or $[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots)$ or $2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= \sqrt{192} = 8\sqrt{3} = 13.85\dots)$		5	M1
	$[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe eg } \sqrt{8^2 - (4\sqrt{3})^2} (= 4)$ And 1 of $[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots) \text{ oe or}$ $\sqrt{8^2 - "4"'^2} (= 4\sqrt{3} = 6.928\dots) \text{ or}$ $2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= 8\sqrt{3} = 13.85\dots)$			M1 [(JM =) 4 + 4√3 implies M2]
	$[AM =]\sqrt{12^2 + ("4\sqrt{3}"')^2} (= \sqrt{192} = 8\sqrt{3} = 13.856\dots) \text{ oe}$ [where M is the midpoint of GH]			M1 Clear intention to be AM (not EC)
	$\tan MAJ = \left(\frac{"4" + "4\sqrt{3}"'}{8\sqrt{3}} \right) \text{ oe eg } \tan MAJ = \left(\frac{10.928\dots}{13.856\dots} \right)$ if student uses sin or cos, then AJ = 17.647... to award marks this must come from a correct method or be correct			M1
	Correct answer scores full marks (unless from obvious incorrect working)	38.3		A1 Accept 38.1 – 38.4 If no marks scored then award SCB2 for $\tan MAJ = \frac{4+x}{\sqrt{x^2+12^2}}$ or SCB1 for $AM = \sqrt{x^2+12^2}$
Total 5 marks				

11 (b)	$(UR =)42 \tan 30 (= 14\sqrt{3} = 24.2(487\dots)) \text{ or}$ $(UR =) \frac{42 \times \sin 30}{\sin (90-30)} (= 14\sqrt{3} = 24.2(487\dots))$		3	M1
	$\tan (UMR) = \left(\frac{"24.248\dots"}{42 \div 2} \right) \text{ or}$ $\tan (UMR) = \left(\frac{"24.248\dots"}{21} \right) \text{ or}$ $\tan (UMR) = \left(\frac{"14\sqrt{3}"'}{21} \right) \text{ or}$ $(UM =) \sqrt{\left(\frac{42}{2} \right)^2 + ("14\sqrt{3}"')^2} (= 7\sqrt{21} = 32.0(780\dots))$ and $\sin (UMR) = \left(\frac{"14\sqrt{3}"'}{7\sqrt{21}} \right) \text{ or } \cos (UMR) = \left(\frac{21}{7\sqrt{21}} \right)$			M1
	Correct answer scores full marks (unless from obvious incorrect working)	49.1		A1 awrt 49.1

12	$9^2 = 11^2 + 16^2 - 2 \times 11 \times 16 \times \cos BCA$ oe or $11^2 = 9^2 + 16^2 - 2 \times 9 \times 16 \times \cos BAC$ or $16^2 = 9^2 + 11^2 - 2 \times 9 \times 11 \times \cos ABC$ or (area of $\triangle ABC = \frac{1}{2} \times 18 \times 2 \times 7 \times 9 (= 47.6235...)$) oe	5	M1	For a start to the correct method to find angle BCA or angle BAC or angle ABC or a fully correct method to find the area of the triangle
	$(\cos BCA = \left(\frac{11^2 + 16^2 - 9^2}{2 \times 11 \times 16} \right)) (BCA = 32.763...) \text{ or}$ $(\cos BAC = \left(\frac{9^2 + 16^2 - 11^2}{2 \times 9 \times 16} \right)) (BAC = 41.409...) \text{ or}$ $(\cos ABC = \left(\frac{9^2 + 11^2 - 16^2}{2 \times 9 \times 11} \right)) (ABC = 105.826...) \text{ or}$ $\frac{1}{2} \times 16 \times BD = "47.6235..."$		M1	For a correct rearrangement for $\cos BCA$ or $\cos BAC$ or $\cos ABC$ or a correct equation to find BD (accept angles to the nearest whole number rounded or truncated as long as not from incorrect working)
	$(BD =) 11 \sin "32.763..." (= 5.95...) \text{ oe eg}$ $11 \sin (180 - "41.4..." - 105.8...) (= 5.95...) \text{ or}$ $9 \sin "41.4..." (= 5.95...) \text{ oe or}$ $\frac{"47.6235..." \times 2}{16} (= 5.95...) \text{ oe or } \sqrt{11^2 - "9.25"} \text{ or } \sqrt{9^2 - "6.75"}^2$ $11 \sin \left(\sin^{-1} \left(\frac{9 \sin "105.826..." }{16} \right) \right) (= 5.95...) \text{ oe}$		M1	For a correct calculation that will lead to the value of BD "47.6235..." may also come from $0.5 \times 9 \times 11 \times \sin "105.8..."$ or $0.5 \times 9 \times 16 \times \sin "41.4..."$ or $0.5 \times 16 \times 11 \times \sin "32.7..."$ [Students may find an angle by sine rule after already finding an angle and use this]
	$\tan FDB = \frac{10}{"5.95..."} \text{ oe}$		M1	For a correct expression for the required angle (in form $\tan x = ...$ or $\cos x = ...$ or $\sin x = ...$) oe
	Correct answer scores full marks (unless from obvious incorrect working)	59.2	A1	awrt 59.2
	SEE OVER FOR ALTERNATIVE SCHEME			Total 5 marks

Angle $DBC = 57.237...$ Angle $ABD = 48.591...$ $AD = 6.75 \text{ m}$ $CD = 9.25 \text{ m}$				
12	$BD^2 = 11^2 - (16 - y)^2$ and $BD^2 = 9^2 - y^2$ oe	5	M1	For 2 different expressions in the same single variable for BD or BD^2
	$11^2 - (16 - y)^2 = 9^2 - y^2$ ($y = 6.75$ or $x = 9.25$)		M1	Equating the 2 expressions
	$BD = \sqrt{9^2 - (16 - "9.25")^2}$ or $\sqrt{11^2 - "9.25"}^2$ ($= 5.95$)		M1	A correct calculation to find BD ("9.25" or "6.75" must come from a correct method)
	$\tan FDB = \frac{10}{"5.95..."} \text{ oe}$		M1	For a correct expression for the required angle (in form $\tan x = ...$ or $\cos x = ...$ or $\sin x = ...$) oe
	Correct answer scores full marks (unless from obvious incorrect working)	59.2	A1	awrt 59.2
				Total 5 marks